

Title: Phase Shift Keying

Objectives:

1. To implement Phase Shift Keying (PSK) digital modulation technique (transmitter) 2.
- To write a program for PSK in MATLAB

OUTCOME: analyze different band pass digital transmission and reception of signal

Draw Circuit Diagram:

Procedure:

- Assemble the circuit as shown in circuit diagram
- Apply digital modulating signal and analog carrier signals with given frequency and amplitude.
- Observe the PSK output on DSO and tabulate the readings.
- Draw the waveforms of modulating, carrier and PSK signal.
- Write a program in MATLAB to observe the PSK waveforms.

Observation Table:

Modulating Signal		Carrier Signal		BFSK Signal		
Amplitude	Frequency	Amplitude	Frequency 1	Amplitude	Phase Shift for logic 0	Phase Shift for logic 1
4.24V	214Hz	4.8V	1kHz	8.8V	40.8°	174°

Matlab Code:

```

clc;
clear;
close all;
%% Input Binary Data
data = [1 0 1 1 0 0 1 0];
%% Parameters
A = 1;
fc = 10;           % Carrier frequency
phi1 = 0;          % Phase for Logic 1
phi0 = pi;         % Phase for Logic 0 (180 degrees)
Tb = 1;            % Bit duration
fs = 1000;         % Sampling frequency
t = 0:1/fs:Tb-1/fs;
%% Generate NRZ Signal

```



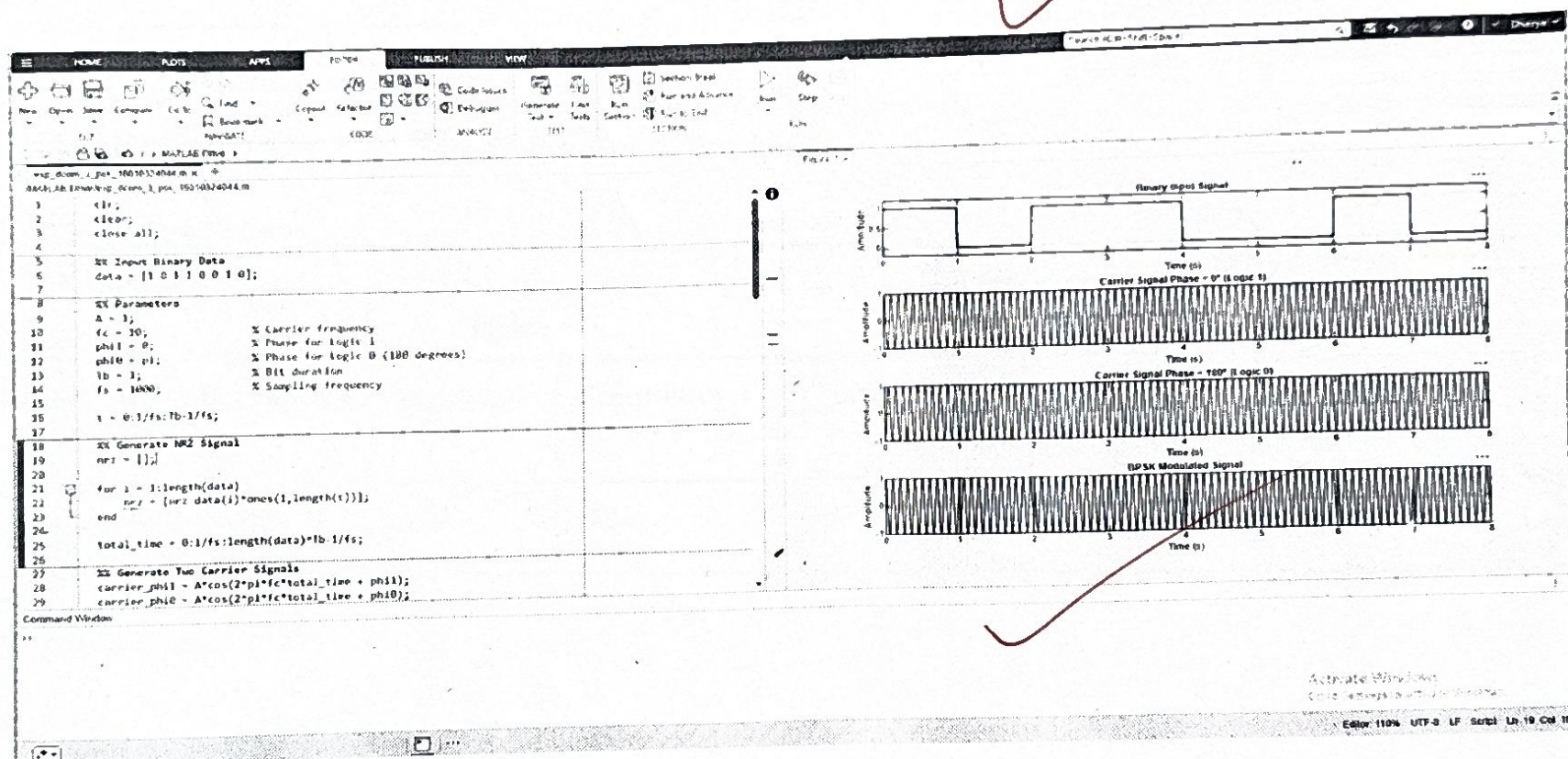
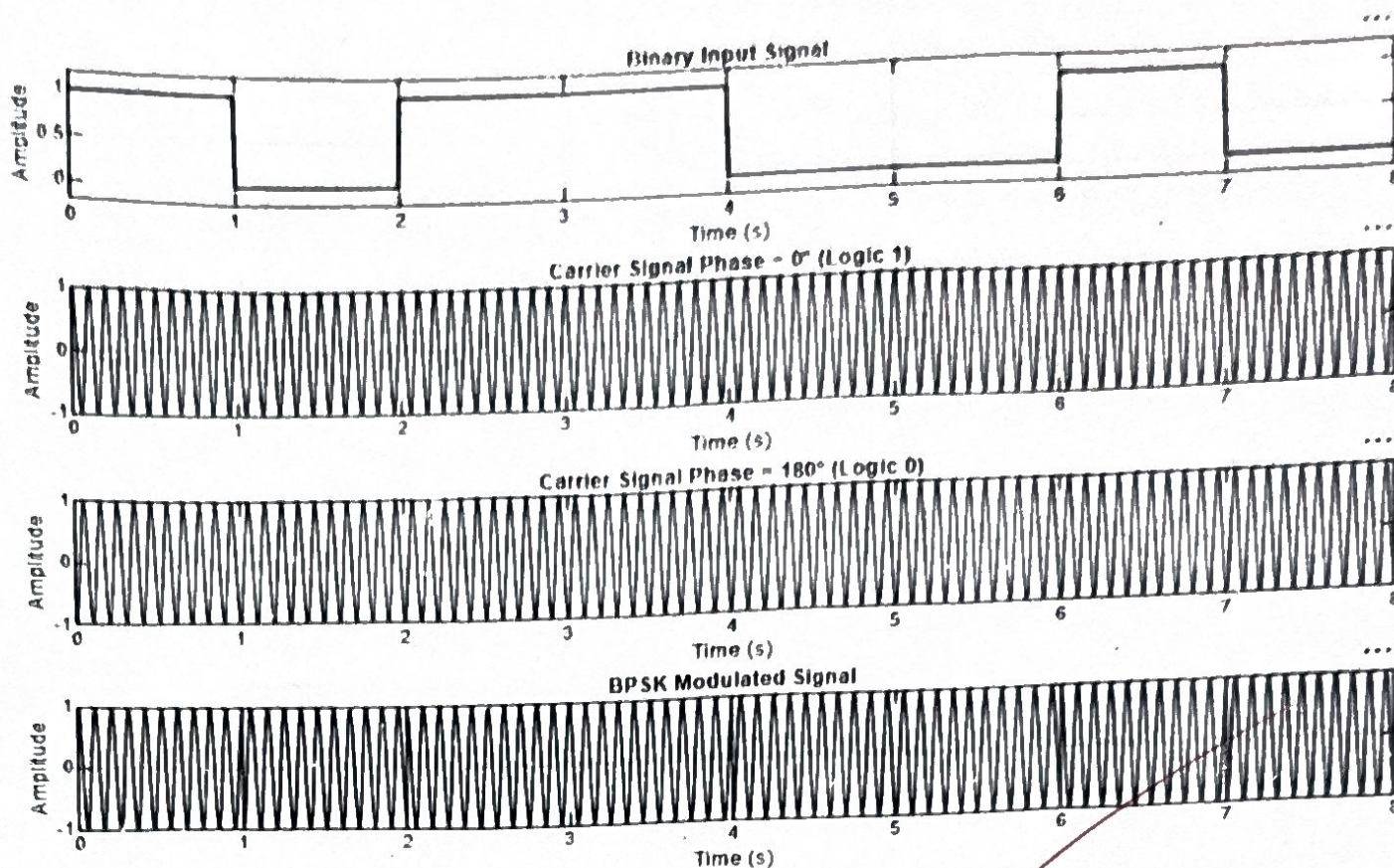
```

nrz = [];
for i = 1:length(data)
    nrz = [nrz data(i)*ones(1,length(t))];
end
total_time = 0:1/fs:length(data)*Tb-1/fs;
%% Generate Two Carrier Signals
carrier_phi1 = A*cos(2*pi*fc*total_time + phi1);
carrier_phi0 = A*cos(2*pi*fc*total_time + phi0);
%% Generate PSK Signal
psk = [];
for i = 1:length(data)
    if data(i) == 1
        % Logic 1 → Phase = 0 degrees
        psk = [psk A*cos(2*pi*fc*t + phi1)];
    else
        % Logic 0 → Phase = 180 degrees
        psk = [psk A*cos(2*pi*fc*t + phi0)];
    end
end
%% Plot Signals
figure;
% 1. Binary Input
subplot(4,1,1)
plot(total_time,nrz,'LineWidth',2)
axis([0 length(data) -0.2 1.2])
grid on
title('Binary Input Signal')
xlabel('Time (s)')
ylabel('Amplitude')
% 2. Carrier Phase 0
subplot(4,1,2)
plot(total_time,carrier_phi1,'LineWidth',1.2)
grid on
title('Carrier Signal Phase = 0° (Logic 1)')
xlabel('Time (s)')
ylabel('Amplitude')
% 3. Carrier Phase 180
subplot(4,1,3)
plot(total_time,carrier_phi0,'LineWidth',1.2)
grid on
title('Carrier Signal Phase = 180° (Logic 0)')
xlabel('Time (s)')
ylabel('Amplitude')
% 4. PSK Signal
subplot(4,1,4)
plot(total_time,psk,'LineWidth',1.5)
grid on
title('BPSK Modulated Signal')
xlabel('Time (s)')
ylabel('Amplitude')

```


Matlab Output:

Figure 1



Discussion/Conclusion:

In this experiment, I successfully implemented PSK modulation and the PSK waveform was observed on DSO and spectrum analyzer. It was verified that two different carrier phases are used to represent Logic 0 and Logic 1. The PSK waveform was also generated using MATLAB and the results were verified.

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SCALE :-

x-axis: Any

2 Unit = 2V

y-axis: Unit Period / $T_c = 1 \text{ unit}$

Modulating Signal

$+V = 4V$

T_c

Sq Wave: 214 Hz

T_c

Carrier

Signal

$+V = 4V$

T_c

Sine Wave: 1 kHz

BPSK

Signal

$+V = 8V$

T_c

Modulated Wave

